

- ④ replace two more variables production in right hand with new production
- ⑤ replace other ~~the~~ illegal production "aB"

⑦ Greibach Normal Form

A CFG is Greibach Normal Form if :

$$\begin{cases} A \rightarrow b \\ A \rightarrow b C_1 C_2 \dots C_n \end{cases} \quad \text{where } A, C_1, C_2, \dots$$

is Non-Terminal Symbol

Steps :

- 1) Remove Unit Productions and Null Productions
- 2) check whether CFG has in chomsky Normal Form (CNF) and convert it to CNF if it is not
- 3) change the names of the Non-Terminal Symbol into some A_i in ascending order of i

order
of simplify CFG:

1. Null Production
2. Unit Production
3. Non-generating Symbols
4. Unreachable Symbols

Examples $S \rightarrow CA|BB$
 $B \rightarrow b|SB$
 $C \rightarrow b$

$\nearrow A \rightarrow a$

already in Chomsky Normal Form

$\Rightarrow$ Replace with: S with A_1

B with A_4

C with A_2

A with A_3

$\Rightarrow A_1 \rightarrow A_2 A_3 | A_4 A_4$

$A_4 \rightarrow b | A_1 A_4$

$A_2 \rightarrow b$

$A_3 \rightarrow a$

Step 4. Alter the Rules so that the Non-Terminals are in ascending order such that, If the Production is of the form $A_i \rightarrow A_j x$, then, $i < j$ and should never be $i \geq j$

$$A_1 \rightarrow A_2 A_3 \mid A_4 A_4$$

$$\textcircled{1} A_4 \rightarrow b \mid \underline{A_1} A_4$$

$$\Rightarrow A_4 \rightarrow b \mid \underline{A_2 A_3} A_4 \mid A_4 A_4 A_4$$

$$\Rightarrow A_4 \rightarrow b \mid b A_3 A_4 \mid \underline{A_4 A_4 A_4}$$

Left Recursion

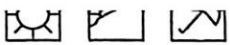
Step 5: Remove Left Recursion

~~Q~~ Q2 GNF - Remove Left Recursion

add $A_4 \rightarrow b \mid b A_3 A_4 \mid A_4 A_4 A_4$

$$z \rightarrow A_4 A_4 z \mid A_4 A_4$$

$$\Rightarrow A_4 \rightarrow b \mid b A_3 A_4 \mid b z \mid b A_3 A_4 z$$



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Now the grammar is:

$$A_1 \rightarrow A_2 A_3 / A_4 A_4$$

$$A_4 \rightarrow b / b A_3 A_4 / b z / b A_3 A_4 z$$

$$z \rightarrow A_4 A_4 / A_4 A_4 z$$

$$A_2 \rightarrow b$$

$$A_3 \rightarrow a$$

$\Rightarrow$

$$A_1 \rightarrow b A_3 / b A_4 / b A_3 A_4 A_4 / b z A_4 /$$

$$b A_3 A_4 z A_4$$

$$A_4 \rightarrow b / b A_3 A_4 / b z / b A_3 A_4 z$$

$$z \rightarrow b A_4 / b A_3 A_4 A_4 / b z A_4 / b A_3 A_4 z A_4 /$$

$$b A_4 z / b A_3 A_4 A_4 z / b z A_4 A_4 z / \cancel{b z} b A_3 A_4 z A_4 z$$

$$A_2 \rightarrow b$$

$$A_3 \rightarrow a$$

Order of Greibach Normal Form

① Remove null production & unit production

② Convert to CNF (Chomsky)

③ Rename Non-Terminal symbol to A_i in increasing order of i



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④ If there are $A_i \rightarrow A_j X$, substitute A_j to A_i make sure $i < j$.

⑤ Remove left Recursive by substitution

$$A \rightarrow A\alpha \mid \beta$$

$$\text{add } B \rightarrow \alpha \mid \alpha\beta$$

左递归 $\rightarrow$ 右递归

$$\text{then } A \rightarrow \beta B \mid \beta$$

(82) Pumping Lemma For CFG

pumping (for CFL) is used to prove that a language is NOT Context Free

! If A is a Context Free Language, then, A has a Pumping Length " P " such that any string $|s| > P$ may be divided into 5